

Assignment 4: GPU and Cooling System Theory

Task 1: Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

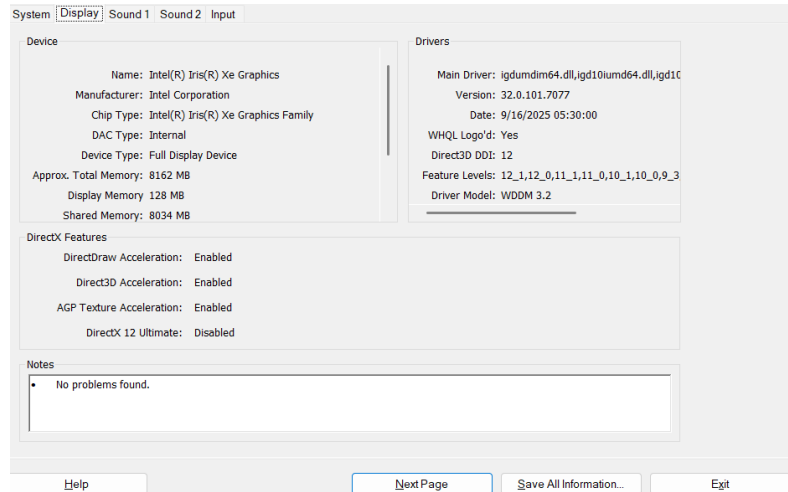


Fig 1. GPU

Task 2: Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

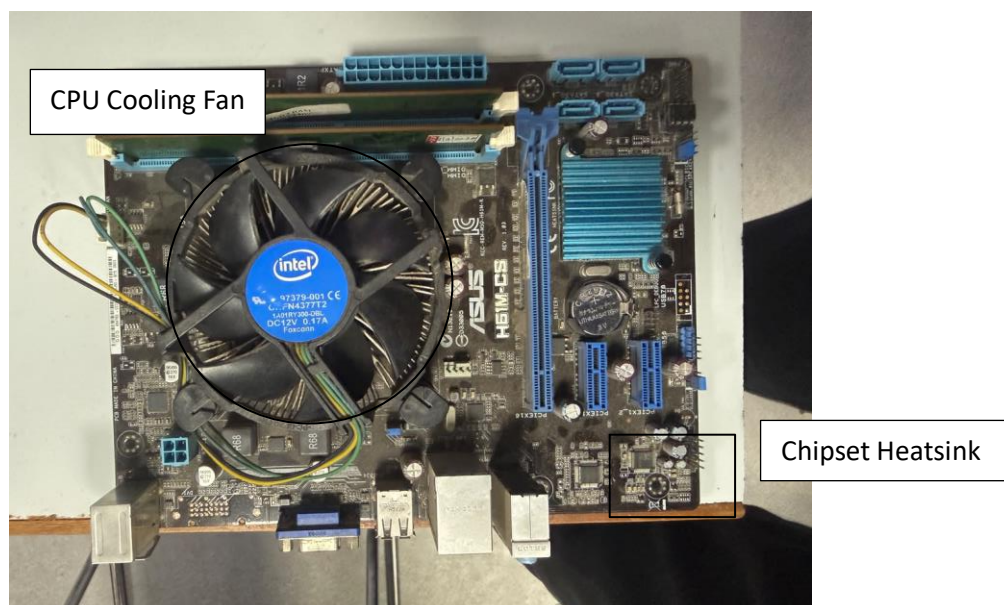


Fig. 2. Cooling System

Main Cooling components and it's work

CPU Fan: Pulls cool air through the heatsink fins and pushes hot air away, preventing the processor from overheating

CPU Cooling Fan: Absorbs heat from the CPU and transfers it to the surrounding air, where the fan dissipates it.

CPU Chipset Heatsink: Cools the motherboard chipset (Intel H61), helping maintain stable communication between the CPU, memory, storage, and expansion devices.

Task 3: Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Table 1. Comparison between Air Cooling GPU and Liquid Cooling GPU

Air Cooling (GPU)	Liquid Cooling (GPU)
Uses fans and metal heatsinks as cooling method.	Uses coolant, water blocks, pumps, radiator, and fans to remove heat.
Good for normal gaming and everyday uses.	Excellent for high-performance gaming, overclocking, and workstation tasks.
Lower cost and more affordable.	Higher cost due to additional components.
Very low maintenance; mainly cleaning dust from fans and heatsinks.	Required periodic maintenance such as checking coolant levels, tubing, and leaks.
Can become noisy under heavy GPU load because fans spin faster.	Usually quieter since heat is dissipated more efficiently and fans often run at lower speed

Task 4: Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

Answer: I would choose **air cooling** for my gaming PC. Since I will mainly be streaming IPL matches and playing graphics-heavy games, a high-quality air-cooled GPU provides excellent performance while being more affordable than liquid cooling. It is also easier to install and requires very little maintenance, making it a reliable choice for long gaming sessions. Unless I plan to do extreme overclocking, air cooling offers the best balance of cost, cooling performance, and reliability.